

Deva Anand

(413) 315-1715 | devaanand@umass.edu | devaanand.com | linkedin.com/in/devaaa | github.com/Deva-1903

Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, TypeScript, JavaScript, C++, Java, SQL, Bash, HTML/CSS

LLM & Agentic Systems

LLM agents, tool use, provider routing/fallback, prompt engineering, LLM evaluation, OpenAI / Groq APIs, Ollama

ML & NLP

PyTorch, HuggingFace Transformers, NLP, RAG, retrieval / reranking, NumPy

DevOps & Observability

CI/CD, GitHub Actions, Docker, pipeline instrumentation, logging / monitoring, incident triage, Spark UI profiling

Web & Cloud

React, FastAPI, Node.js, Express, REST APIs, async services, Azure, AWS, GCP, Git

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Selected for a measured **GitHub Copilot productivity pilot**, routing tech-debt PRs, code migrations, and security-finding triage through **AI-assisted workflows** to reduce manual engineering effort; maintained a versioned `copilot-instructions.md` conventions file to keep AI-generated code aligned with codebase patterns.
- Owned backend reliability across a **multi-tenant NHS eTriage platform (8+ UK clients, 10,000+ monthly submissions)**: monitored production logs during feature releases and client integrations, investigated and shipped fixes for tenant-specific incidents, and hardened authentication and rate limiting.
- Architected a full-stack internal tool (React, Node.js, MongoDB) used daily by the AI team to label data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual work by **100+ hours/month** and improved throughput by **60%**.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Integrated **LLMs** into a chat product to generate context-aware replies from conversation history and user input, iterating on **prompt design** and response-quality heuristics, while building **20+ Node.js + Fastify REST APIs** over PostgreSQL.

Projects

AgenticSearch - AI Agent with Output Verification & Provenance

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage **AI agent** (Python/FastAPI) that turns noisy, open-ended inputs into structured, verified output via typed retrieval planning, rerank, deterministic extraction with LLM fallback, and an **output-verification stage**, returning cell-level provenance for every value.
- Hardened the service with **150+ automated tests**, provider routing/fallback across LLM backends, and pipeline instrumentation, validating and debugging **AI-generated output** before it reaches the user.

cf-ai-research-scout - Real-Time AI Assistant on Event-Driven Infra

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, Durable Objects, WorkflowEntrypoint, WebSockets, D1

- Built an event-driven **AI assistant**: an agent fetches sources, runs multi-step analysis, and **streams responses over WebSocket** in real time, an interactive chat-style tool deployed end-to-end on serverless infrastructure.
- Designed **durable workflow boundaries** so multi-step tasks survive process restarts and individual tool failures, with agent state persisted server-side, enabling reliable long-running automation without an external database.

Spark ETL Optimization - Distributed Pipeline Diagnostics

github.com/Deva-1903/Spark-ETL-Optimization

Python, Apache Spark / PySpark, Spark UI, SQL

- Profiled stage-level timings and telemetry in the **Spark UI** to localize shuffle and skew bottlenecks across a distributed ETL pipeline on a **~1.4B-row** dataset, then documented a repeatable diagnostic playbook.
- Cut end-to-end runtime **~25%** (17 to 19 min down to 14.3 min) and reduced the analytics tail-latency ratio (P99/P50) from **2.14x to 1.74x** via broadcast joins, Adaptive Query Execution, skew-join handling, and partition/column pruning.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.